

1. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52

$$1a) \left[\text{Mean} = \sum_{i=1}^n \frac{w_i x_i}{w_i} \right] \quad \frac{1}{n} \sum_{i=1}^n \frac{w_i x_i}{w_i}$$

$$= \frac{809}{27}$$

$$= 29.96 = 30$$

$$\text{Median} = \frac{n+1}{2}$$

$$= \frac{27+1}{2}$$

$$= 14^{\text{th}} = 25$$

$$\text{Median} = \underline{\underline{25}}$$

b) Mode and modality

Frequency

25 and 35 are repeating for 4 times it is

bimodel

$$(c) \text{ Midrange} = \frac{\text{minimum} + \text{maximum}}{2}$$

$$= \frac{13 + 70}{2}$$

$$= 41.5$$

$$\text{Midrange} = 41.5$$

d) First Quartile (Q_1) and Third Quartile Q_3

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

Q_1 and Q_3

Lower half

middle value = 20

$$Q_1 = 20$$

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

upper half

middle value = 35

$$Q_3 = 35$$

$$IQR = 15$$

$$Q_3 - Q_1$$

e) Five Number Summary

min, Q_1 , median, Q_3 , max

$$\bullet \text{ minimum} = 13$$

$$\bullet Q_1 = 20$$

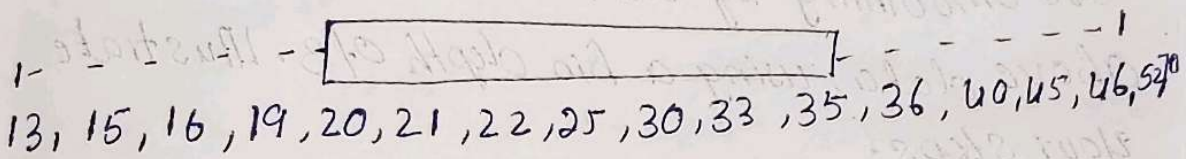
$$\bullet \text{ median} = 25$$

$$\bullet Q_3 = 35$$

$$\bullet \text{ maximum} = 70$$

f. Boxplot of the data

13, 15, 16, 19, 20, 21, 22, 25, 30, 33, 35, 36, 40
45, 46, 52, 70



2. Given the following measurements for the variable

age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28

standardize the variable by the following
compute the mean absolute deviation of age

Given data

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

$$\text{Mean } \bar{x} = \frac{330}{10} = 33$$

Age $(x - \bar{x})$

$$18 - 15$$

$$22 - 11$$

$$25 - 8$$

$$42 - 9$$

$$28 - 5$$

$$43 - 10$$

$$33 - 0$$

$$35 - 2$$

$$56 - 23$$

$$28 - 5$$

$$\text{MAD} = \frac{88}{10} = 8.8$$

3. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order)

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25,

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 66, 52, 70

Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

Total values = 27

no. bin

depth = 3

Total no. of bins = $\frac{27}{3} = 9$ bins required

Bin	values	Bin mean	Smoothed value
B ₁	13, 15, 16	14.67	14.67, 14.67, 14.67
B ₂	16, 19, 20	18.33	18.33, 18.33, 18.33
B ₃	20, 21, 22	21.00	21.00, 21.00, 21.00
B ₄	22, 25, 25	24.00	24.00, 24.00, 24.00
B ₅	25, 25, 30	26.67	26.67, 26.67, 26.67
B ₆	33, 33, 35	33.67	33.67, 33.67, 33.67
B ₇	35, 35, 35	35.00	35.00, 35.00, 35.00
B ₈	36, 40, 45	40.33	40.33, 40.33, 40.33
B ₉	46, 52, 70	56.00	56.00, 56.00, 56.00

4. Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000 (a) min-max normalization by setting $\min = 0$ and $\max = 1$ (b) z-score normalization

200, 300, 400, 600, 1000

b) min-max

$$x' = \frac{x - \min}{\max - \min}$$

$$\min = 200$$

$$\max = 1000$$

Value

$$\begin{aligned} \text{Normal value} \\ (200 - 200) / 800 \\ = 0.00 \end{aligned}$$

200

$$\frac{300 - 200}{800} = 0.125$$

300

$$(400 - 200) / 800 = 0.250$$

400

$$(600 - 200) / 800 = 0.500$$

600

$$(1000 - 200) / 800 = 1.000$$

1000

b)
$$z = \frac{x - \mu}{\sigma}$$

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = 500$$

Value

$(x - \mu)$

$(x - \mu)^2$

200

-300

1.061

300

-200

0.707

400

-100

0.354

600

-100

0.354

1000

-500

1.768

5. Suppose a group of 12 sales price were

has been sorted as follows: 5, 10, 11, 13, 15,

50, 55, 72, 92, 204, 215. Partition them into the

following methods. (a)